

Numerical Computation and Machine Learning Basics

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Special Lectures for Machine Learning

Outline

Numerical Computation

Machine Learning Basics: Learning Algorithms

Machine Learning Basics: Capacity, Overfitting and Underfitting

Machine Learning Basics: Hyperparameters and Validation Sets

Machine Learning Basics: Estimator, Bias, and Variance

Machine Learning Basics: Maximum Likelihood Estimator and
Bayesian Statistics

Machine Learning Basics: Various Methodologies and Challenges
Motivating Deep Learning

Numerical Computation I

1. Overflow and Underflow

- ▶ Basic limit: Computers use only finite bytes in calculating numerical values in various operations (rounding error/quantization).
- ▶ Underflow: Nearly 0 becomes 0 (log operation can fail).
- ▶ Overflow: Going to infinite.

Numerical Computation II

- For an example, considering

$$\text{softmax}(\mathbf{x})_i = \frac{\exp(x_i)}{\sum_{j=1}^n \exp(x_j)}.$$

1. When $\min_j x_j$ becomes larger, overflow can occur.
2. When $\max_j x_j$ becomes smaller toward negative infinite, underflow can occur.
3. One of easy solution to this case: $x_j \leftarrow x_j - \min_j x_j$.

Numerical Computation III

- Poor conditioning.

$$f(\mathbf{x}) = \mathbf{A}^{-1}\mathbf{x}, \mathbf{A} \in \mathbb{R}^{n \times n}$$

The eigenvalues of $\mathbf{A}$ are denoted by λ_j s, then *condition number* is $\max_{i,j} |\frac{\lambda_i}{\lambda_j}|$.

- The larger condition number value implies the sensitivity to input variation (error), not a computation problem.

Numerical Computation IV

2. Gradient-Based Optimization

- ▶ Basically, machine learning minimizes the object function.
- ▶ Object function, loss or error function, cost function. In this lecture, we'll ignore the subtle differences between various terminologies.
- ▶ Note:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0)$$

- Originated at $\mathbf{x}_0$, if the direction of $\mathbf{x} - \mathbf{x}_0$ is equal to $-\nabla f(\mathbf{x}_0)$, the $f(\mathbf{x})$ becomes smaller than $f(\mathbf{x}_0)$.

Numerical Computation V

- Types of critical points $\nabla f(\mathbf{x}) = 0$: Minimum, maximum, and saddle point.

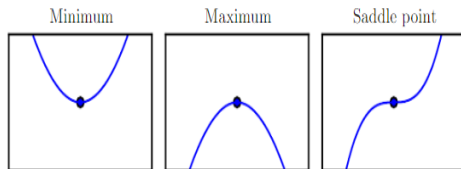


Figure 1: Three types of critical points.

Numerical Computation VI

- Global minima and local minima.

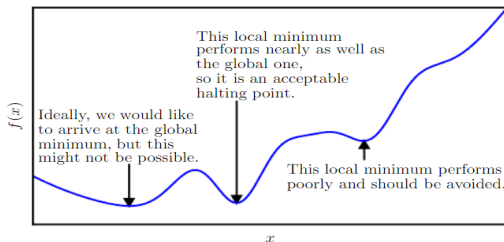


Figure 2: Global minima and local minima.

Numerical Computation VII

- ▶ Usually, we consider the object function such that $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
- ▶ Partial derivative $\frac{\partial}{\partial x_i} f(\mathbf{x})$ measures how f changes as only the variable x_i changes.
 - Gradient $\nabla_{\mathbf{x}} f(\mathbf{x})$ is the vector of partial derivatives.

Numerical Computation VIII

- Finding the direction of minimizing object function.

$$\begin{aligned} & \min_{\mathbf{u}, \mathbf{u}^T \mathbf{u} = 1} \mathbf{u}^T \nabla_{\mathbf{x}} f(\mathbf{x}) \\ &= \min_{\mathbf{u}, \mathbf{u}^T \mathbf{u} = 1} \|\mathbf{u}\| \|\nabla_{\mathbf{x}} f(\mathbf{x})\| \cos \theta \end{aligned}$$

- *Steepest descent* proposes a new point.

$$\mathbf{x}' = \mathbf{x} - \epsilon \nabla_{\mathbf{x}} f(\mathbf{x})$$

1. The ϵ is referred to as the learning rate (observing the $f(\mathbf{x} - \epsilon \nabla_{\mathbf{x}} f(\mathbf{x}))$).
2. For several values of ϵ , choose the one that results in the smallest objective function value. This last strategy is called a line search.
3. Ascending an objective function of discrete parameters is called "hill climbing."

Numerical Computation IX

3. Jacobian and Hessian

- ▶ Negative gradient is the direction of the non-increasing object function.
- ▶ When we consider the changes of elements of the gradient at each update, IT CAN BE IRREGULAR.
- ▶ Using the change of gradient can be required to have a better optimizer.
- ▶ Jacobian matrix for a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$.

$$J_{i,j} = \frac{\partial f(\mathbf{x})_i}{\partial x_j}$$

Numerical Computation X

- Hessian matrix: a derivative of a derivative.

$$H(f)(\mathbf{x})_{i,j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$$

- We can think of the second derivative as measuring curvature (Hessian is the Jacobian of the gradient).

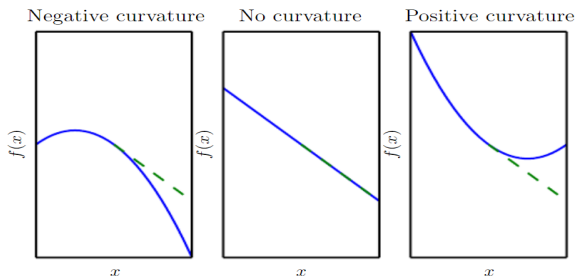


Figure 3: Various curvatures.

Numerical Computation XI

- ▶ Continuous and commutative: $\mathbf{H}(f)(\mathbf{x})_{i,j} = \mathbf{H}(f)(\mathbf{x})_{j,i}$.
- ▶ Real and symmetric, and an eigendecomposition can be adopted.

$$\begin{aligned}f(\mathbf{x}) &= f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{g} + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}_0) \\f(\mathbf{x}_0 - \epsilon \mathbf{g}) &= f(\mathbf{x}_0) - \epsilon \mathbf{g}^\top \mathbf{g} + \frac{1}{2}\epsilon^2 \mathbf{g}^\top \mathbf{H} \mathbf{g}\end{aligned}$$

- ▶ Role of Hessian.

1. Optimal ϵ is

$$\frac{\mathbf{g}^\top \mathbf{g}}{\mathbf{g}^\top \mathbf{H} \mathbf{g}}.$$

2. Second derivative test: identifying the local minimum, local maxima, and saddle point.

Numerical Computation XII

- Local minimum
 $\mathbf{u}^T \mathbf{H} \mathbf{u}$ is positive for all $\mathbf{u}$ (positive definite $\mathbf{H}$).
- Local maximum
 $\mathbf{u}^T \mathbf{H} \mathbf{u}$ is negative for all $\mathbf{u}$ (negative definite $\mathbf{H}$).
- Saddle point
 $\mathbf{u}^T \mathbf{H} \mathbf{u}$ is negative/positive for some $\mathbf{u}$.

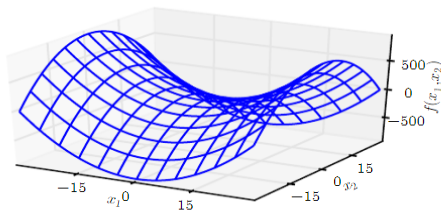


Figure 4: Example of a saddle point.

Numerical Computation XIII

- ▶ Note that the change of gradient w.r.t. each coordinate can be different.
- ▶ Newton's method (second order approximation).

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^T \nabla_{\mathbf{x}} f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^T \mathbf{H}(f)(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$$

$$\mathbf{x}^* = \mathbf{x}_0 - \mathbf{H}(f)(\mathbf{x}_0)^{-1} \nabla_{\mathbf{x}} f(\mathbf{x}_0)$$

Numerical Computation XIV

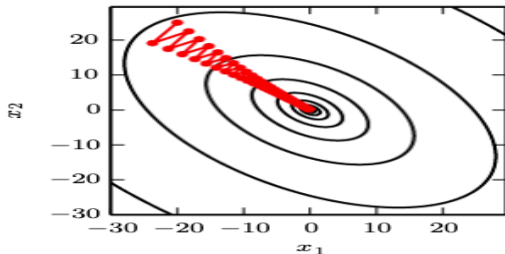


Figure 5: Path of gradient descent.

- Restriction on a function f , *Lipschitz continuous*.

convex, hessian 등 계산 어렵지만
이것만은 된다!!

$$\forall \mathbf{x}, \mathbf{y} \quad |f(\mathbf{x}) - f(\mathbf{y})| \leq \mathcal{L} \|\mathbf{x} - \mathbf{y}\|.$$

Numerical Computation XV

- Ideas from the analysis of convex optimization algorithms can be helpful in proving the convergence of deep learning algorithms, but in general, the importance of convex optimization is greatly diminished in the context of deep learning.

Numerical Computation XVI

4. Constrained Optimization

$$\arg \min_{\mathbf{x}} f(\mathbf{x}) \text{ subject to } \mathbf{x} \in \mathcal{S}.$$

- ▶ Here, points that lie within the set $\mathcal{S}$ are called feasible points.
- ▶ Simple approach: projection into $\mathcal{S}$ when $\mathbf{x} = \mathbf{x} - \epsilon \nabla_{\mathbf{x}} f(\mathbf{x})$.
- ▶ Structured approach: $f(x, y) = f(\theta)$ where $x = \sin(\theta)$, $y = \cos(\theta)$.
Searching minimizing with a constraint $\sqrt{x^2 + y^2} = 1$.

Numerical Computation XVII

- ▶ Karush–Kuhn–Tucker(KKT) approach provides a very general solution to constrained optimization.
- ▶ Setup.
 1. $\mathcal{S} = \{\forall i, g(\mathbf{x}) = 0, \forall j, h(\mathbf{x}) \leq 0\}$, equality and inequality regulations.
 2. Object: $\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$.
 3. Generalized Lagrangian.

$$L(\mathbf{x}, \lambda, \alpha) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^n \alpha_j h_j(\mathbf{x})$$

where $\lambda_i > 0$ and $\alpha_j \geq 0$.

Numerical Computation XVIII

- ▶ Goal: $\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = \min_{\mathbf{x}} \max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha})$.
 1. The above is $f(\mathbf{x})$ when $\mathbf{x} \in \mathcal{S}$.
 2. The above is infinite when $\mathbf{x} \notin \mathcal{S}$.
 3. If the $\mathbf{x}^*$ is the feasible optimizer and $f(\mathbf{x}^*) \neq \infty$,
$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = \min_{\mathbf{x}} \max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha})$$
$$(\sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^n \alpha_j h_j(\mathbf{x}) = 0).$$
- ▶ Temporarily, we let h_i is active when $h_i(\mathbf{x}) = 0$. If the h_i is not active, α_i should be 0.

$$\boldsymbol{\alpha} \odot \mathbf{h} = \mathbf{0}$$

Numerical Computation XIX

- ▶ Necessary conditions for the optimal $\mathbf{x}^*$ (not ensuring uniqueness) when the convexity of f and g holds and the h satisfies the Slater's condition (Karush, 1939; Kuhn and Tucker, 1951).
 1. The gradient of generalized Lagrangian w.r.t. $\mathbf{x}$ is $\mathbf{0}$.
 2. All constraints on both $\mathbf{x}$ and the KKT multipliers are satisfied.
 3. “complementary slackness”: $\alpha \odot \mathbf{h}(\mathbf{x}) = \mathbf{0}$.

Machine Learning Basics: Learning Algorithms I

1. Tasks, Performance Measure, and Experience

- ▶ Classification: $f : \mathbb{R}^n \rightarrow \{0, \dots, K\}$ via class probabilities.
- ▶ Regression: $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
- ▶ Transcription: For example, audio waveform or emits $\rightarrow$ text of codes.
- ▶ Machine Translation: For example, French to English.
- ▶ Structured Output: For example, image captioning or trees for the natural language.

Machine Learning Basics: Learning Algorithms II

- ▶ Anomaly detection: Finding irregular or uncharacteristic behaviors.
- ▶ De-noising: Corrupted $\tilde{x} \in \mathbb{R}^n$ to clean $x \in \mathbb{R}^n$.
- ▶ Density or probability mass function estimation: Learn $p : \mathbb{R}^n \rightarrow \mathbb{R}$.
- ▶ Performance measure
 1. Accuracy: portion of correct prediction (in classification-type tasks).
 2. Mean squared error: Mean of the squared difference between the predicted and true (in regression-type tasks).
 3. The choice of performance measure may seem straightforward and objective: For example, density function estimation or transcription.

Machine Learning Basics: Learning Algorithms III

► Experience

1. Most of the learning algorithms in this book can be understood as being allowed to experience an entire *dataset*. An element of dataset is called a data point.
 2. Broadly categorized into supervised and unsupervised based on the experience allowed to have during the learning process.
-
- Supervised learning: Experiencing a dataset containing many features, then learning useful properties of the structure of this dataset (clustering, density estimation, and others).
 - Unsupervised learning: Experiencing a dataset containing features, but each example is also associated with a label or target (classification, regression, and others).
 - Semi-supervised learning / Self-supervised learning.

Machine Learning Basics: Learning Algorithms IV

- ▶ Examples of probability of (un)supervised learning.

$$p(\mathbf{x}) = \prod_{i=1}^n p(x_i \mid x_{i-1}, \dots, x_1)$$
$$p(y \mid \mathbf{x}) = \frac{p(\mathbf{x}, y)}{\sum_{y'} p(\mathbf{x}, y')}$$

- ▶ Dataset is denoted by $\{\mathbf{x}_i, y_i\}_{i=1}^N$, and design matrix is

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} \in \mathbb{R}^{n \times N}.$$

Machine Learning Basics: Capacity, Overfitting and Underfitting I

1. Generalization Error

- ▶ Generalization: The ability to perform well on previously unobserved inputs.
- ▶ Training error: Error on the training dataset.
- ▶ Test error: Error on the test dataset where the test dataset is separated from the training dataset used in learning.

Machine Learning Basics: Capacity, Overfitting and Underfitting II

- An example of regression.

$$\frac{1}{m^{\text{train}}} \| \mathbf{X}^{\text{train}} \mathbf{w} - \mathbf{y} \|^2$$
$$\frac{1}{m^{\text{test}}} \| \mathbf{X}^{\text{test}} \mathbf{w} - \mathbf{y} \|^2$$

- Statistical interpretation.
 1. Usually, assume that the training and test set follows the same distribution with i.i.d. data generating regime.
 2. The expected training error of a randomly selected model is equal to the expected test error of that model.

Machine Learning Basics: Capacity, Overfitting and Underfitting III

- ▶ We sample the training set, then use it to choose the parameters to reduce training set error, and then sample the test set.
- ▶ The performance of the machine learning algorithm is validated by
 1. Making the training error small.
 2. Make the gap between training and test error small.
- ▶ Underfitting and Overfitting.
 1. The training model cannot achieve the lower training error.
 2. The gap between training and test error is large.

Machine Learning Basics: Capacity, Overfitting and Underfitting IV

- ▶ Solutions to avoid these problems.
 1. Altering the model capacity (informally, a model's capacity is its ability to fit a wide variety of functions).
 2. Using hypothesis space (the set of functions learning algorithm can select as the solution).

Machine Learning Basics: Capacity, Overfitting and Underfitting V

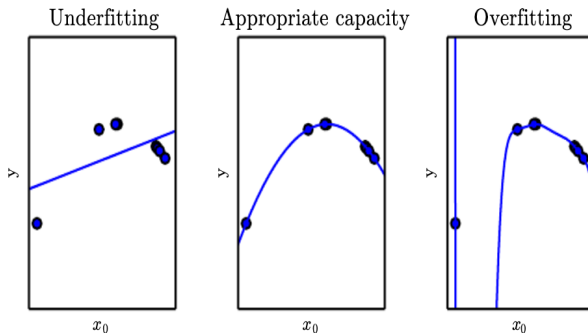


Figure 6: Hypotheses space becomes larger in polynomial regressions.

Machine Learning Basics: Capacity, Overfitting and Underfitting VI

- ▶ *Representational capacity* of the model: Discriminated from the *effective capacity* by reducing the training error. Usually higher than the *effective capacity*.
- ▶ Important concepts.
 1. Ocam's (1287–1347) Razor.

Among competing hypotheses that explain known observations equally well, we should choose the "simplest" one.
 2. Vapnik-Chervonenkis dimension.

Measuring the capacity of a binary classifier, the largest possible value of m for which there exists a training set of m different x points that the classifier can label arbitrarily.

Machine Learning Basics: Capacity, Overfitting and Underfitting VII

► Implication of VC.

The gap between the training and test error can be reduced when the sample size increases and be larger when the model complexity increases.

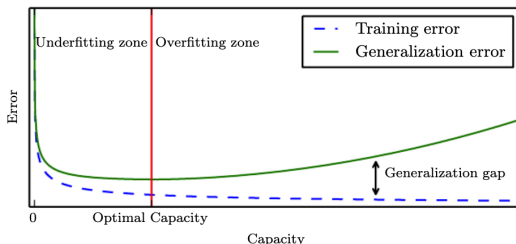


Figure 7: Gap along with the model capacity.

Machine Learning Basics: Capacity, Overfitting and Underfitting VIII

- ▶ Deep networks.
 - Less theoretical understanding for the measure capacity: not sharp bound for the model complexity and non-optimal algorithm for the representational capacity.
- ▶ Nonparametric models.
 - Nearest neighborhood.
 - Nested models.
 - Same structure with increasing parameters.

Machine Learning Basics: Capacity, Overfitting and Underfitting IX

- ▶ Bayes error: The error incurred by an oracle making predictions from the true distribution $p(\mathbf{x}, y)$.
- ▶ Various situations.
 1. Lower capacity and large gap.
 2. High capacity and large gap.
 3. The gap depends on datasets.
 4. Large data makes the gap smaller.

Machine Learning Basics: Capacity, Overfitting and Underfitting X

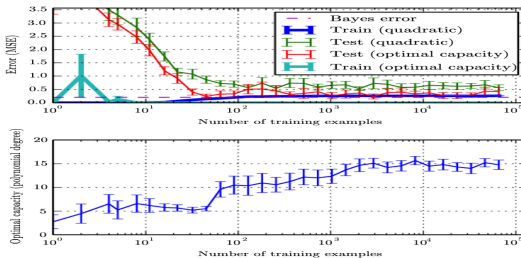


Figure 8: Training and test error as sample size increases.

Machine Learning Basics: Capacity, Overfitting and Underfitting XI

2. No Free Lunch

- ▶ Basically, the learning algorithm cannot experience all possible sets. Learning the underlying probability structure for better generalization.
- ▶ However, it has limitations.

No Free Lunch Theorem

Averaged over all possible data-generating distributions, every classification algorithm (including the random guess) has the same error rate when classifying previously unobserved points.

Machine Learning Basics: Capacity, Overfitting and Underfitting XII

- ▶ We cannot construct a better universal algorithm for all distributions, requiring what kinds of distributions are relevant to the "real world".
- ▶ Regularization and prior of Bayesian arises.

3. Regularization

- ▶ We should build a set of preferences into the learning algorithm inspired by the "No Free Lunch Theorem".
- ▶ Controlling the size of hypotheses is a simple method. Preference for some functions in the hypotheses space can be considered.

Machine Learning Basics: Capacity, Overfitting and Underfitting XIII

- ▶ Weight decay: $J(\mathbf{w}) = \text{MSE}_{\text{train}}(\mathbf{w}) + \lambda \Omega(\mathbf{w})$ where $\Omega(\mathbf{w}) = \mathbf{w}^T \mathbf{w}$.
- ▶ Def. of *Regularization*: Any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.

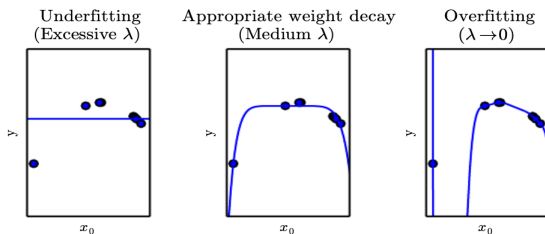


Figure 9: Regularization effect with polynomial degree and weight decay.

Machine Learning Basics: Hyperparameters and Validation Sets I

1. Validation

- ▶ Hyperparameter.
 1. Capacity hyperparameter.
 2. Optimization hyperparameter.
- ▶ If learned on the training set, such hyperparameters would always choose the maximum possible model capacity, resulting in overfitting.
- ▶ To find a better hyperparameter for the generalization, a validation set is required.

Machine Learning Basics: Hyperparameters and Validation Sets II

- ▶ Typically, one uses about 80 percent of the training data for training and 20 percent for validation.
- ▶ Since the validation set is used to "train" the hyperparameters, the validation set error will underestimate the generalization error.
- ▶ After all hyperparameter optimization is complete, the generalization error may be estimated using the test set.

Machine Learning Basics: Hyperparameters and Validation Sets III

2. Cross Validation

- ▶ When the test (validation) set is small, it can be problematic, such as hundreds or thousands of sizes.
- ▶ Dividing the data randomly into k -folds and using the one fold to test (validate) the training on other folds.
- ▶ All folds are assigned to the test (validation) set, and all errors' average or standard deviation is calculated.

Machine Learning Basics: Estimator, Bias, and Variance I

1. Point Estimation

- ▶ Assume that we have parameters to be estimated.
- ▶ Two types of point: vector θ and function f (can be parametrized by w).

$$\hat{\theta}_m = g(\mathbf{x}_1, \dots, \mathbf{x}_m)$$

- ▶ We can use the empirical risk minimization approach such as $\hat{\theta} = \arg \min_{\theta} L(\theta \{\mathbf{x}_i, y_i\}_{i=1}^m)$ where L is an object function for the learning.

Machine Learning Basics: Estimator, Bias, and Variance II

2. Bias and Variance

- ▶ Bias $\text{bias}(\hat{\theta}_m) = \mathbb{E}[\hat{\theta}_m] - \theta$.
- ▶ Asymptotically unbiased if $\lim_{m \rightarrow \infty} \text{bias}(\hat{\theta}_m) = 0$.
- ▶ Standard Error.
 1. $\text{Var}(\hat{\theta}_m)$.
 2. $\text{SE}(\hat{\theta}_m) = \sqrt{\text{Var}(\hat{\theta}_m)}$.
 3. The SE assesses the uncertainty of confidence for the estimator, which is used for the confidence interval.

Machine Learning Basics: Estimator, Bias, and Variance III

- ▶ Trading off bias and variance to minimize MSE for one-dimensional θ .

Decomposition of MSE.

$$\mathbb{E}[|\hat{\theta}_m - \theta|^2] = |\mathbb{E}[\hat{\theta}_m] - \theta|^2 + \text{Var}(\theta)$$

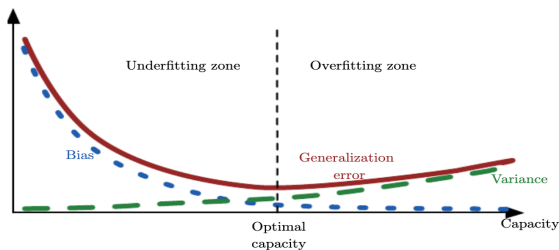


Figure 10: Bias and variance trade-off.

Machine Learning Basics: Estimator, Bias, and Variance IV

3. Consistency

- ▶ Mathematical formulation.

$$\text{plim}_{m \rightarrow \infty} \hat{\theta}_m \rightarrow \theta$$

- ▶ Interpretation in prob.

$$\forall \epsilon > 0, \mathbb{P}(|\hat{\theta}_m - \theta| > \epsilon) \rightarrow 0 \text{ as } m \rightarrow \infty.$$

Machine Learning Basics: Machine Learning Basics: Maximum Likelihood Estimator and Bayesian Statistics I

1. Maximum Likelihood Estimator

- ▶ Having a long history, and the basic principle is how the probability of data can be large when the parameterized probability varies.
- ▶ Maximum likelihood estimator θ_{ML} .

$$\begin{aligned}\arg \max_{\theta} p(\mathbf{x}_1, \dots, \mathbf{x}_m \mid \theta) &= \arg \max_{\theta} \prod_{i=1}^m p(\mathbf{x}_i \mid \theta) \\ \theta_{\text{ML}} &= \arg \max_{\theta} \sum_{i=1}^m \log p(\mathbf{x}_i \mid \theta)\end{aligned}$$

Machine Learning Basics: Machine Learning Basics:

Maximum Likelihood Estimator and Bayesian Statistics II

- ▶ When considering the empirical distribution such that $\hat{p}_{\text{data}} = \sum_{i=1}^m \delta_{\mathbf{x}_i}$, we have

$$\theta_{ML} = \arg \max_{\theta} \mathbb{E}_{\hat{p}_{\text{data}}} [\log p(\mathbf{x} \mid \theta)].$$

- ▶ Interpretation w.r.t. KL divergence.

$$\text{KL}(\hat{p}_{\text{data}} \parallel p) = \mathbb{E}_{\mathbf{x} \sim \hat{p}_{\text{data}}} \left[\underbrace{\log \hat{p}_{\text{data}}(\mathbf{x}) - \log p(\mathbf{x})}_{\text{Minimizing}} \right]$$

Machine Learning Basics: Machine Learning Basics: Maximum Likelihood Estimator and Bayesian Statistics III

- ▶ The part of conditional likelihood is omitted (HW).

Machine Learning Basics: Machine Learning Basics:

Maximum Likelihood Estimator and Bayesian Statistics IV

► Properties of MLE

1. The true distribution p must lie within the model family $p(\cdot | \theta)$ (coverage).
2. The data-generating distribution p must correspond to exactly one value of θ (identifiability).

Machine Learning Basics: Machine Learning Basics: Maximum Likelihood Estimator and Bayesian Statistics V

2. Bayesian Statistics

- ▶ Considering θ as not unknown but *random*.
- ▶ Based on the Bayesian, prior distribution $p(\theta)$ arise.
- ▶ Prior can be uniform or non-informative to strengthen the data dependency.

Machine Learning Basics: Machine Learning Basics:

Maximum Likelihood Estimator and Bayesian Statistics VI

- ▶ Bayesian's framework is knowing θ by observing data, *posterior*

$$p(\theta \mid \mathbf{x}_1, \dots, \mathbf{x}_m) = \frac{p(\mathbf{x}_1, \dots, \mathbf{x}_m \mid \theta)}{p(\mathbf{x}_1, \dots, \mathbf{x}_m)}$$

- ▶ Prediction can be done more elegantly under the assumption of conditional independence.

$$p(\mathbf{x} \mid \mathbf{x}_1, \dots, \mathbf{x}_m) = \int p(\mathbf{x} \mid \theta) p(\theta \mid \mathbf{x}_1, \dots, \mathbf{x}_m) d\theta.$$

Machine Learning Basics: Machine Learning Basics:

Maximum Likelihood Estimator and Bayesian Statistics VII

► Comparison of MLE and Bayesian

1. Model: $y \sim \mathcal{N}(\mathbf{X}\mathbf{w}, I)$.
2. MLE of $\mathbf{w}$.

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

3. Posterior of $\mathbf{w}$ when $p(\mathbf{w}) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Lambda}_0)$.

$$\mathcal{N}(\boldsymbol{\mu}_m, \boldsymbol{\Lambda}_m)$$

where

$$\begin{aligned}\boldsymbol{\mu}_m &= \boldsymbol{\Lambda}_m(\mathbf{X}^\top \mathbf{y} + \boldsymbol{\Lambda}_0^{-1} \boldsymbol{\mu}_0) \\ \boldsymbol{\Lambda}_m &= (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Lambda}_0)^{-1}.\end{aligned}$$

Machine Learning Basics: Machine Learning Basics:

Maximum Likelihood Estimator and Bayesian Statistics VIII

4. Usually, we let $\mu_0 = \mathbf{0}$, and $\mathbf{\Lambda}_0 = \frac{1}{\alpha} \mathbf{I}$. Here, α is a hyperparameter concerning the effect of prior belief.
5. Prior belief is related to a penalty term such as $\alpha \mathbf{w}^T \mathbf{w}$.

► MAP (Maximum a Posteriori).

$$\theta_{\text{MAP}} = \arg \max_{\theta} \left\{ \sum_{i=1}^m \log p(\mathbf{x}_i | \theta) + \log p(\theta) \right\}.$$

Machine Learning Basics: Various Methodologies and Challenges Motivating Deep Learning I

1. Various Methodologies

► Supervised learning.

1. Probabilistic supervised learning: regression, logistic regression, and generalized linear model.
2. Support vector machine.
3. Decision tree, and k -nearest neighborhood.

► Unsupervised learning.

1. Principle component analysis.
2. The k -means clustering.

Machine Learning Basics: Various Methodologies and Challenges Motivating Deep Learning II

2. Stochastic Gradient Descent

$$\theta \leftarrow \theta - \epsilon \mathbf{g},$$

where $\mathbf{g} = \frac{1}{m'} \nabla_{\theta} \sum_{i=1}^{m'} L(\mathbf{x}_i, y_i \mid \theta)$.

- Note that we use the (disjoint) mini-batch samples (sub-sampled from the whole dataset) for each update.

Machine Learning Basics: Various Methodologies and Challenges Motivating Deep Learning III

- For example, if the whole dataset has 100 samples and the size of the mini-batch is 10, then ten updates will be done using the whole sample, which is called one epoch. Usually, we use tens, hundreds, and thousands of epochs.
- Usually, different mini-batches are used between epochs due to the random subsampling.

Machine Learning Basics: Various Methodologies and Challenges Motivating Deep Learning IV

3. Challenges Motivating Deep Learning

- Curse of dimension: The required samples filling out the input space densely grows exponentially.

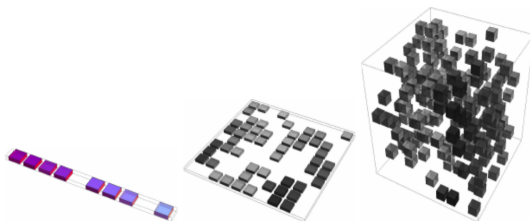


Figure 11: Curse of dimension ("Figure graciously provided by Nicolas Chapados" in the textbook).

Machine Learning Basics: Various Methodologies and Challenges Motivating Deep Learning V

- ▶ Local constancy and smoothness regularization.
 1. Most widely used of these implicit "priors" is smoothness or local constancy prior.
 2. Basic idea is that the dense countable points can approximate the function, such that $f(\mathbf{x} + \epsilon) \approx f(\mathbf{x})$.
- ▶ The nearest neighborhood provides the function at most m different values. The kernel methods can be flexible but also assume the local constancy or almost linear.
- ▶ In general, to distinguish $O(k)$ regions in input space, all these methods, including the decision tree, require $O(k)$ examples.

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- ▶ Whether it is possible to represent a complicated function efficiently, and whether it is possible for the estimated function to generalize well to new inputs.
- ▶ The key insight is that a very large number of regions, such as $O(2k)$, can be defined with $O(k)$ samples (by the deep layers and many nodes).
- ▶ Basic concept is the *composition of factors* from simple to complex). The exponential advantages conferred by the use of deep distributed representations counter the exponential challenges posed by the curse of dimensionality.

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4. Manifold Learning

- ▶ A manifold is a connected region, a set of points associated with a neighborhood around each point.
- ▶ Approximated well by considering only a small number of degrees of freedom (dimension), embedded in a high dimensional space.

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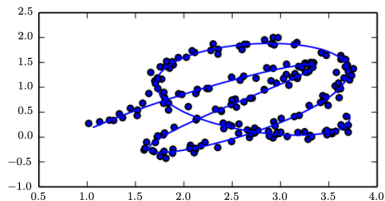


Figure 5.11: Data sampled from a distribution in a two-dimensional space that is actually concentrated near a one-dimensional manifold, like a twisted string. The solid line indicates the underlying manifold that the learner should infer.

Figure 12: Example of a manifold.

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- ▶ Manifold learning assumes that the data lies along a low-dimensional manifold.
- ▶ Preference to manifold learning
 1. In favor of the manifold hypothesis, the probability distribution over images, text strings, and sounds that occur in real life is highly concentrated.
 2. We can imagine such neighborhoods and transformations, at least informally (rotation, color, and others).
 3. In everyday life, we can think of roads as 1-D manifolds embedded in 3-D space.